

Chapter 3: Data Representation Methods in the Computer System

In this unit, you will learn about:

- Computer data representation.
- Decimal, Binary, Octal, and Hexadecimal number systems.
- Most and Least Significant Positional Values (MSB/LSB).
- Conversions between decimal, binary, octal, and hexadecimal systems.
- Data storage capacity and coding systems (ASCII, BCD, EBCDIC, Unicode).

3.1 Computer Data Representation

Computers represent data through electronic signals. When a key like "A" is pressed on the keyboard, it is converted into an electronic symbol (pattern of 0s and 1s) that the CPU can understand. After processing, the character is displayed on the screen.

3.1.1 Number Systems

A group of numbers that a computer can understand is called a "Number System." The value of a digit depends on the position it occupies within the number.

Number System	Base Value	Digits Used
Binary	2	0, 1
Octal	8	0, 1, 2, 3, 4, 5, 6, 7
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Hexadecimal	16	0-9, A, B, C, D, E, F

3.1.2 Binary Numbers in Data Representation

Computers use two signal states: high voltage (1) and low voltage (0), corresponding to "On" and "Off." All data (text, images, sound) is represented using these two digits.

3.2 Types of Number Systems

3.2.1 Decimal Number System (Base 10)

Used in daily life. Digits: 0-9. Positional values are powers of 10 (e.g., 10^0 , 10^1 , 10^2).

Example: 25_{10}

$$25 = (2 \times 10^1) + (5 \times 10^0) = 20 + 5$$

3.2.2 Binary Number System (Base 2)

Consists of digits 0 and 1. Positional values are powers of 2 (e.g., 2^0 , 2^1 , 2^2).

Bit: The smallest unit of data in a computer (0 or 1).

3.2.3 Octal Number System (Base 8)

Uses eight digits: 0-7. Positional values are powers of 8 (e.g., 8^0 , 8^1 , 8^2).

3.2.4 Hexadecimal Number System (Base 16)

Uses 0-9 and A-F (where A=10, B=11, C=12, D=13, E=14, F=15). It is often used to represent large binary numbers in a shorter format.

3.3 Most and Least Significant Digits

- **MSD / MSB:** Most Significant Digit/Bit (leftmost digit with the highest value).

- **LSD / LSB:** Least Significant Digit/Bit (rightmost digit with the lowest value).

3.4 Conversions

3.4.1 Decimal to Binary/Octal/Hexadecimal

Divide the decimal number by the target base (2, 8, or 16) repeatedly until the quotient is 0. The remainders, read from bottom to top, form the new number.

Example: Convert 12_{10} to Binary

$12 / 2 = 6$, remainder 0

$6 / 2 = 3$, remainder 0

$3 / 2 = 1$, remainder 1

$1 / 2 = 0$, remainder 1

Result: 1100_2

3.4.2 Binary/Octal/Hexadecimal to Decimal

Multiply each digit by its positional weight (base raised to the power of its position) and sum the results.

Example: Convert 1101_2 to Decimal

$(1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$

$= 8 + 4 + 0 + 1 = 13_{10}$

3.5 Data Storage Capacity

Data is measured in bits and bytes. **1 Byte = 8 Bits.**

Unit	Symbol	Value
Kilobyte	KB	1024 Bytes (2^{10})

Megabyte	MB	1024 KB (2^{20})
Gigabyte	GB	1024 MB (2^{30})
Terabyte	TB	1024 GB (2^{40})

3.6 Coding Systems

- **BCD (Binary Coded Decimal):** Uses 4 bits to represent each decimal digit (0-9).
- **ASCII (American Standard Code for Information Interchange):** Originally 7 bits (128 characters). Used for standard text.
- **EBCDIC (Extended Binary Coded Decimal Interchange Code):** 8 bits (256 characters). Used mainly in IBM mainframes.
- **Unicode:** 16 bits (65,536 characters). Can represent characters from almost all world languages (Sinhala, Tamil, Japanese, etc.).

Summary

- Computers process data in binary form (0 and 1).
- Number systems like Octal and Hexadecimal are used to simplify binary representation for humans.
- Conversions allow us to move between human-readable decimal and machine-understandable binary.
- Coding systems ensure that text, images, and sound can be digitally represented and exchanged.

Activity Questions

1. Convert 83_{10} to a binary number.
2. Convert 10110111_2 to an octal number.
3. Convert $23D_{16}$ to a binary number.
4. If "A" is represented as 1000001 in ASCII, what is the code for "F"?